

MAT 4953 01F

SP: Optimal Control

Course Syllabus - Summer 2026



Course Information

Course Description: Prerequisite: Consent of instructor. An organized course offering the opportunity for specialized study not normally or not often available as part of the regular course offerings. May be repeated for credit when the topics vary, but not more than 6 semester credit hours, regardless of discipline, will apply to a bachelor's degree. Generally offered: Fall, Spring, Summer. This course has Differential Tuition. Course fee: DL01 \$75.

Credit Hours: 3

Course Modality: Online only, at set time

Meeting Times

Duration: 05/26/2026-06/30/2026

Campus: Internet

Location: [INT](#)

Time(s): MTWRF 4:15 PM - 5:30 PM

Learning Goals

At the end of the course, you will be able to:

- This course provides an introductory, but thorough, development of the mathematical aspects of optimal control theory and calculus of variations with an insight into application of optimal control theory to scientific and engineering models.

Communicate with Me

Instructor Name: Mostafa Fazly

Department
Mathematics

Student Hours
4:15-5:30. Before or after lectures. By appointment.

Email Address: mostafa.fazly@utsa.edu

Preferred Method of Communication
Canvas Inbox. Or Email.

About Me & My Teaching Philosophy

Dr. Mo Fazly

Assistant Dean for Faculty Support, College of Sciences,
Associate Professor, Department of Mathematics,
University of Texas at San Antonio.

<https://orcid.org/0000-0001-7290-2140>

Research Interests: Differential Equations and Applied Mathematics

Course Materials

Lecture Notes

Authors: Instructor

Additional Information

Will be provided

Calculus of Variations and Optimal Control Theory; A Concise Introduction

Authors: Daniel Liberzon

Publisher: Princeton University Press (2012)

Required/Recommended/Optional: Optional

Optimal Control Applied to Biological Models,

Authors: Suzanne Lenhart and John T. Workman,

Publisher: Chapman and Hall/CRC (2007)

Required/Recommended/Optional: Recommended

Dynamic Optimization: The Calculus of Variations and Optimal Control in Economics and Management.

Authors: Morton I. Kamien and Nancy L. Schwartz

Publisher: North-Holland, New York, (1991)

Required/Recommended/Optional: Optional

An Introduction to Mathematical Optimal Control Theory

Authors: Lawrence C. Evans,

Required/Recommended/Optional: Optional

Additional Information

Link: <https://math.berkeley.edu/~evans/control.course.pdf>

To enhance your learning and adequately prepare for all assignments and assessments, please consider all course materials listed above, along with additional content provided in class, posted to our course site, or included in the Course Schedule, as required reading and study unless indicated as optional.

Additional Course Information

Prerequisites: MAT 3613 or equivalent

The course topics will be covered according to the following plan, if time permits. This may be updated during the semester if necessary.

Topics:

Basic Optimal Control Problems

The Basic Problem and Necessary Conditions

Optimality Condition, Adjoint Equation, Transversality Condition

Interpretation of the Adjoint

Bang-Bang Control and Applications

Pontryagin's Maximum Principle; Statement, Ideas, Applications

Existence and Uniqueness Results

Principle of Optimality

The Hamiltonian and Autonomous Problems

Importance of Convexity

Examples of Variational Problems, Brachistochrone Problem and Catenary Problem

The Fundamental Problem.

Basic Calculus of Variations Problems, Euler-Lagrange Equation

Calculus of Variations Versus Optimal Control

Extremals. The Insufficiency of Extremality

Fundamental Lemma of Calculus of Variations

Extremal versus Minimizer

Necessary Conditions and Special Cases

Review

Assessments and Assignments

Four Take-Home Midterm Tests (each 20 percent) and Final Exam/Project (20 percent). See Canvas for more information.

There will be four midterm exams and a final exam. Homework problems include analytical work and computational work. The number of homework assignments and deadlines will be determined by the instructor. There will be no extensions on any due dates. No make up exams.

Activities and Grading

Activity	Quantity	%
Midterm Exams	4	80
Final Exam	1	20
Total		100%

Distribution of Course Assignments, Their Quantity, and Contribution to Final Grade.

Grade Distribution and Letter Grade

Letter Grade	Grade Range
A+	97.5--100
A	95--97.4
A-	90--94.9
B+	87.5--89.9
B	85--87.4
B-	80--84.9
C+	77.5--79.9
C	75--77.4
C-	70--74.9
D+	67.5--69.9
D	65--67.4
D-	60--64.9
F	0--59.9

Grading Policy

Grade Distribution and Letter Grade

Course Guidelines for the Use of Generative Artificial Intelligence

The use of generative artificial intelligence (AI) tools is not permitted for assignments, tests, exams, or any graded course work.

Course Expectations & Policies

The syllabus is subject to change at the instructor's discretion.

Any changes/corrections to the course materials, assignment dates, or other updates will be communicated on Canvas.

You are responsible for checking Canvas for corrections and updates to the syllabus and announcements. All lectures will be scheduled on Zoom. Check Canvas for Announcements. Online Only Modality.

Instructor-Initiated Drops

[Optional, but the language should not be changed if included. Remove this language in between the brackets.]

This course uses instructor-initiated drops for students who exceed the absence and/or missed assignment limit. Therefore, up to the last day for students to withdraw from an individual course, **[date]**, you will be dropped for exceeding **[insert number of absences/assignments etc.]**. Students will receive at least one courtesy warning when approaching the absence/missed assignment limit. Notification will be sent using ASAP to the student's email address. A subsequent absence or missed assignment will result in being dropped from the course. Notification of being dropped will also be sent through ASAP to the student's email address. *This drop does not affect enrollment in other courses.* Please consult the [Dropping Courses webpage](#) for further details on the process and appeals.

Video and Audio Recording

[Optional, but the language should not be changed if included. Remove this language in between the brackets.]

As the instructor of this course, I may record meetings and lessons. You are expected to follow appropriate University policies and maintain the security of passwords used to access recorded lectures. Recordings may not be published, reproduced, or shared with those not in the class. If the instructor or a UTSA office plans any other uses for the recordings, consent of the students identifiable in the recordings is required before such use unless an exception is allowed by law. For more information on your privacy and class recordings, review [Student Privacy \(FERPA\) in Virtual Classrooms and Other Educational Recordings](#) and the [Guide to Secure Video Conferencing Tools](#).

Academic Freedom

[Optional, but the language should not be changed if included. Remove this language in between the brackets.]

Academic freedom¹ is a cornerstone of the University. Academic freedom in its teaching aspect is fundamental for the protection of the rights of the teacher in teaching and of the student to freedom in learning.² Each faculty member is entitled to full freedom in the classroom discussing the subject that the faculty member teaches.³ The University of Texas at San Antonio will not penalize or discipline members of the faculty because of their exercise of academic freedom.

Along with this freedom comes responsibility. It is the responsibility of faculty members to ensure that topics taught are related to the classroom subject. Students should be free to take reasoned exception to the data or views offered in any course of study and to reserve judgment about matters of opinion, but they are responsible for learning the content of any course of study for which they are enrolled.⁴ It is not the proper role of the university or any outside agency to attempt to shield individuals from ideas and opinions they find unwelcome, disagreeable, or even deeply offensive.⁵ Engaging with new ideas and perspectives helps students grow intellectually and is beneficial to the educational process.

1. Statement adapted from Texas A&M University's [Syllabus Statement Regarding Academic Freedom](#)
2. 1940 Statement of Principles on Academic Freedom and Tenure
3. Board of Regents Rule 31004; HOP Policy 4.02
4. American Association of University Professors Joint Statement on Rights and Freedoms of Students
5. The Chicago Statement

Our Commitment to Inclusivity

[Optional, but the language may not be changed or altered if included. Remove this language in between the brackets.]

The University of Texas at San Antonio, a Hispanic Serving Institution situated in a global city that has been a crossroads of peoples and cultures for centuries, values diversity and inclusion in all aspects of university life. As an institution expressly founded to advance the education of Mexican Americans and other underserved communities, our university is committed to promoting access for all. UTSA, a premier public research university, fosters academic excellence through a community of dialogue, discovery and innovation that embraces the uniqueness of each voice.

Syllabus Changes

[Optional, but the language should not be changed if included. Remove this language in between the brackets.]

The syllabus is subject to change at the instructor's discretion. Any changes/corrections to the course materials, assignment dates, or other updates will be communicated to the students ahead of time. You are responsible for checking Canvas for corrections or updates to the syllabus.

Course Schedule

For a list of important university-wide dates, review [One Stop's academic calendar](#).

Department Policies & Resources

Ombudsperson

The ombudsperson is an advocate who investigates complaints about a specific section or instructor and attempts to resolve them through mediation. Comments or complaints sent to this person will be brought to the attention of department leadership, who will follow up. If you have questions about the logistics of your class, please contact the instructor.

Contact Ilse Rosales Martinez at ilse.rosalesmartinez@utsa.edu

Tutoring Services

UTSA provides many opportunities for student success, including tutoring services for specific math courses. The list below is subject to change based on availability.

MAT 1023, MAT 1043, MAT 1053, and MAT 1073

- We offer peer-to-peer academic support to students enrolled in a variety of courses. Both drop-in tutoring and one-on-one appointments are available.
- **Location:** [Tomás Rivera Center for Academic Excellence](#) MS 2.02.18 (behind the Rowdy statue)

MAT 1053 and MAT 1073

- Synergistic Targeted Academic Review Sessions (STARS), peer-led, interactive, and group focused sessions which cover traditionally difficult topics. The schedule will be provided by instructors through Canvas.
- **Location:** Determined by STARS tutors

MAT 1093, MAT 1213/1214, and MAT 1223/1224

- The Math Gym is unique in that it's a WeBWork lab designed to offer our STEM core students support from math majors and minors through targeted SI sessions. The schedule will be provided by instructors through Canvas
- **Location:** [Math Gym FLN 03.02.08](#)

Note: Course-specific support for upper-division courses may be available for a particular semester. Please request this information from your instructor.

Essential Student Information

- **Important:** Bookmark and visit the [Common Syllabus Information webpage](#) to find important and valuable resources about counseling services, transitory/minor medical issues,

supplemental instruction, tutoring services, academic success coaching, sexual harassment and sexual misconduct, campus safety and emergency preparedness, and the Roadrunner Creed.

- For technical requirements, support, and resources, visit [Academic Innovation's Student Technical Support](#) page.
- UT San Antonio provides reasonable accommodations to students via [Student Disability Services](#). For more details on eligibility, policies, and requirements, please visit [Applying for Accommodations](#) or call (210) 458-4157.
- Your well-being is important. In addition to academic growth, life as a student can include many emotional experiences and opportunities. Being mindful of your own needs and available support can provide you with the tools to successfully navigate these experiences. Student support for well-being, including [24/7 mental health services](#), can be accessed from the [Well-being at UT San Antonio](#) site for students. If needed, we can schedule a meeting, and I can connect you to resources. You are not alone; I am available to help you find resources that might best meet your needs.
- [Student Assistance Services \(SAS\)](#) is available to help you navigate academic and personal challenges. Campus Advocates in SAS are a central point of contact for students seeking clarity on university processes and policies. They also provide a direct connection to helpful resources and advocacy rooted in care. Campus Advocates offer confidential support and personalized guidance to help students succeed.
- The [Roadrunner Pantry](#) is here to help UT San Antonio students stay healthy and focused by providing free access to groceries, toiletries, and other essentials. With locations at the Main and Downtown campuses, the pantry is organized like a small store and offers items such as canned goods, snacks, and hygiene products. Students with a valid UTSA ID can visit the pantry to pick up what they need. Our goal is to ensure every Roadrunner feels supported and has the resources to succeed. If you're ever short on food or supplies, the Roadrunner Pantry is a welcoming space you can count on.
- Students at UT San Antonio are responsible for ensuring their work is consistent with UT San Antonio's standards for academic integrity. Students should review [Section 203 of the UT San Antonio Student Code of Conduct](#) for appropriate standards of academic integrity.
- UT San Antonio provides numerous services for students from counseling to tutoring to a food pantry. Visit [Student Affairs Programs and Services](#) and [Student Success](#) for more information.
- Visit the [UT San Antonio Libraries and Museums](#) site for access to journals, research tutorials, and tech gear you can borrow and to find your department's librarian.
- Enroll in the [Roadrunner Success Playbook](#), an open-enrollment, self-paced, online hub in Canvas tailored to ensure you have the resources you need to excel at UT San Antonio.
- Follow [Digital Learning Netiquette](#) standards for your online communication activities.